Kshiteesh R Bharadwaj

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Profile Summary

Software Development Engineer with **3.5+ years** architecting low-latency, fault-tolerant distributed control planes using **Modern C++ (C++17/20)**, multithreading, gRPC, and cross-node state synchronization. Proven impact: **68% state-window reduction**, **45% MTTR improvement**, **0 safety incidents**, production-grade observability across large-scale systems.

Education

University of Illinois Urbana-Champaign Aug 2026 – Dec 2027
Master of Computer Science
Coursework: DataMining, Applied Machine Learning, Distributed Systems

RV Institute Of Technology and Management Aug 2019 – Aug 2023
B.E in Information Science and Engineering | Rank #3 *GPA: 9.15/10.0*
Coursework: Machine Learning, Data Structures & Algorithms, Operating Systems, Computer Networks

Experience

Software Development Engineer 2 Aug 2025 – Aug 2026
Nokia – Bangalore, India C++17, Protocol Buffers, Python 3, Redis Cache

- **Architected** 2-of-3 voting safety interlock across 3 distributed entities, eliminating all safety violations and cutting **On-Call reliability incidents from 3 to 0 per quarter**, with observability into vote-state.
- **Unified** live-event and post-restart recovery onto one path, resolving 20-second ambiguity, **reducing MTTR by 45%**.
- Reduced network collisions from **12% to 0.3%** via Distributed backoff protocol with real-time telemetry.
- **Optimized** task scheduling with a priority queue for better Capacity Planning, **cutting latency 38% (1.2s to 740ms)** with High Throughput gains using multithreading.

Software Development Engineer 1 Aug 2023 – Aug 2025
Infinera – Bangalore, India C++, gRPC, NETCONF, RESTCONF

- **Designed** 3-stage Production Grade scan lifecycle for Capacity Planning, **eliminating firmware conflicts by 100%**.
- **Unblocked** HAL dispatch thread from 60s transfers via a 4-worker thread pool, **improving responsiveness 95% (340ms to 16ms)** and enabling High Throughput operations.
- Eliminated duplicate HW commands after Distributed system restarts, cutting API Latency **25%**, via single-write atomicity.
- **Resolved** race condition across Microservices, **cutting Distributed system bugs to zero per release**.

Software Development Intern Feb 2023 – Aug 2023
Infinera – Bangalore, India C++17, Protocol Buffers, Python 3, Bash, Linux

- **Built** Production Grade bidirectional streaming Microservice for telemetry with High Throughput support.
- **Developed** noise-filtering algorithm with observability into signal quality, **improving stability 40%**, validated via **276 TDD-driven tests**, GoogleTest.

Technical Skills

Languages	C/C++, Python, Bash, OOP, Yang
Architecture & Concurrency	Distributed Systems, Control Planes, LLD, High Throughput, Multithreading
Protocols	gRPC (Async/Streaming), Protobuf, NETCONF, RESTCONF
Storage & Networks	Redis (redis plus-plus), Sockets (TCP/IP), IPC, HAL Integration
Testing & Devops	TDD(GTest), RCA, Debugging, Telemetry, Linux, CMake, Git, CI/CD, Docker, Agile

Projects

RateGuard: Distributed Rate Limiter C++, gRPC, Protocol Buffers, Redis Cluster, etcd, Prometheus, Docker

- Built distributed rate limiter reducing API rejection rate by **58%** with Tradeoff Analysis between fairness and throughput.
- **Fixed** race condition causing quota overages by implementing atomic compare-and-swap, **eliminating violations by 100%**, ensuring **On-Call Reliability** across nodes.

Klassy: Cafeteria Management System Node.js, PostgreSQL, TypeScript, WebSocket, Docker, Kubernetes

- **Led** cashless order system adopted college-wide, **reducing transaction time by 60%** via real-time state sync.
- **Delivered** live order tracking with dynamic editing and unique IDs, **eliminating manual errors by 100%**, enabling seamless Feature Delivery and Observability into cafeteria operations.

Flow state: intelligent Task Scheduler Bun, ElysiaJS, React, Vite, SQLite, TypeScript, Capacitor, Whisper AI

- **Architected** scalable backend with intelligent task-scheduling microservice, **reducing query latency by 70%** via optimized database indexing with High Throughput concurrent request handling.
- **Developed** proactive scheduling service with real-time event streaming, state persistence, **enabling 99.5% uptime**.